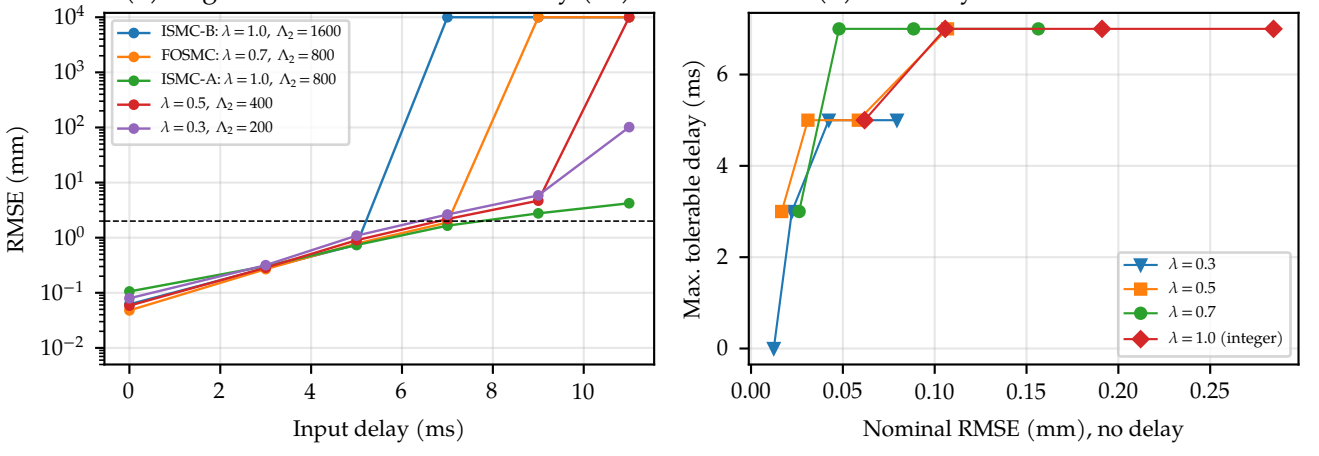
(a) Degradation with actuation delay (S2)
RMSE (mm)
Input delay (ms)
ISMC-B: $\lambda = 1.0$, $\Lambda_2 = 1600$
FOSMC: $\lambda = 0.7$, $\Lambda_2 = 800$
ISMC-A: $\lambda = 1.0$, $\Lambda_2 = 800$
$\lambda = 0.5$, $\Lambda_2 = 400$
$\lambda = 0.3$, $\Lambda_2 = 200$
(b) Accuracy–robustness trade-off
Max. tolerable delay (ms)
Nominal RMSE (mm), no delay
$\lambda = 0.3$
$\lambda = 0.5$
$\lambda = 0.7$
$\lambda = 1.0$ (integer)